

What is a Bidirectional RNN?

A **Bidirectional RNN (BiRNN)** is an extension of the standard RNN.

- A **normal RNN** reads sequences **from past to future** (left → right).
- A **BiRNN** has **two RNNs**:
 - One processes the sequence forward (left → right).
 - The other processes it backward (right → left).
- The outputs from both are combined (concatenated, added, etc.) at each time step.

This means the network has **both past and future context** when making predictions.

Why is this useful in NLP?

Language often requires **both previous and upcoming words** for full understanding.

Example sentence:

"I went to the bank to deposit money."

- If you only see "I went to the bank", *bank* could mean a **riverbank**.
- But "to deposit money" clarifies it's a **financial institution**.

A **BiRNN** sees both contexts and makes better predictions.

Advantages of BiRNN in NLP

1. **Uses Full Context**
 - Forward RNN: past words only.
 - Backward RNN: future words only.
 - BiRNN: combines both → deeper understanding.
2. **Better Accuracy**
 - Helps in tasks where meaning depends on both directions (NER, sentiment, translation).
3. **Disambiguates Meaning**
 - Words like *bank*, *lead*, *bat* need future context to resolve ambiguity.
4. **Improves NLP Tasks**
 - Sentiment analysis: understands negations.
 - POS tagging: predicts tags better using surrounding words.
 - Named Entity Recognition: clarifies entities with forward + backward cues.

Examples in NLP

1. Named Entity Recognition (NER)

Sentence:

"Apple is looking to acquire a London-based startup."

- Forward RNN may think *Apple* = *fruit*.
- Backward RNN sees "*startup*" and realizes *Apple* = *company*.
- BiRNN correctly tags **Apple** → **Organization**.

2. Sentiment Analysis

Sentence:

"I didn't enjoy the movie as much as I expected."

- Forward RNN may see "*enjoy the movie*" and think it's positive.
- Backward RNN catches "*didn't*".
- BiRNN combines both → predicts **Negative sentiment**.

Analogy

- **Unidirectional RNN**: Reading a book only once from start to finish.
- **Bidirectional RNN**: Reading it forward **and** backward → richer understanding.

In summary:

A **Bidirectional RNN** processes text in both directions, capturing **past + future context**.

This improves performance in many NLP tasks like **NER, sentiment analysis, POS tagging, and machine translation**, where meaning depends on both surrounding words.